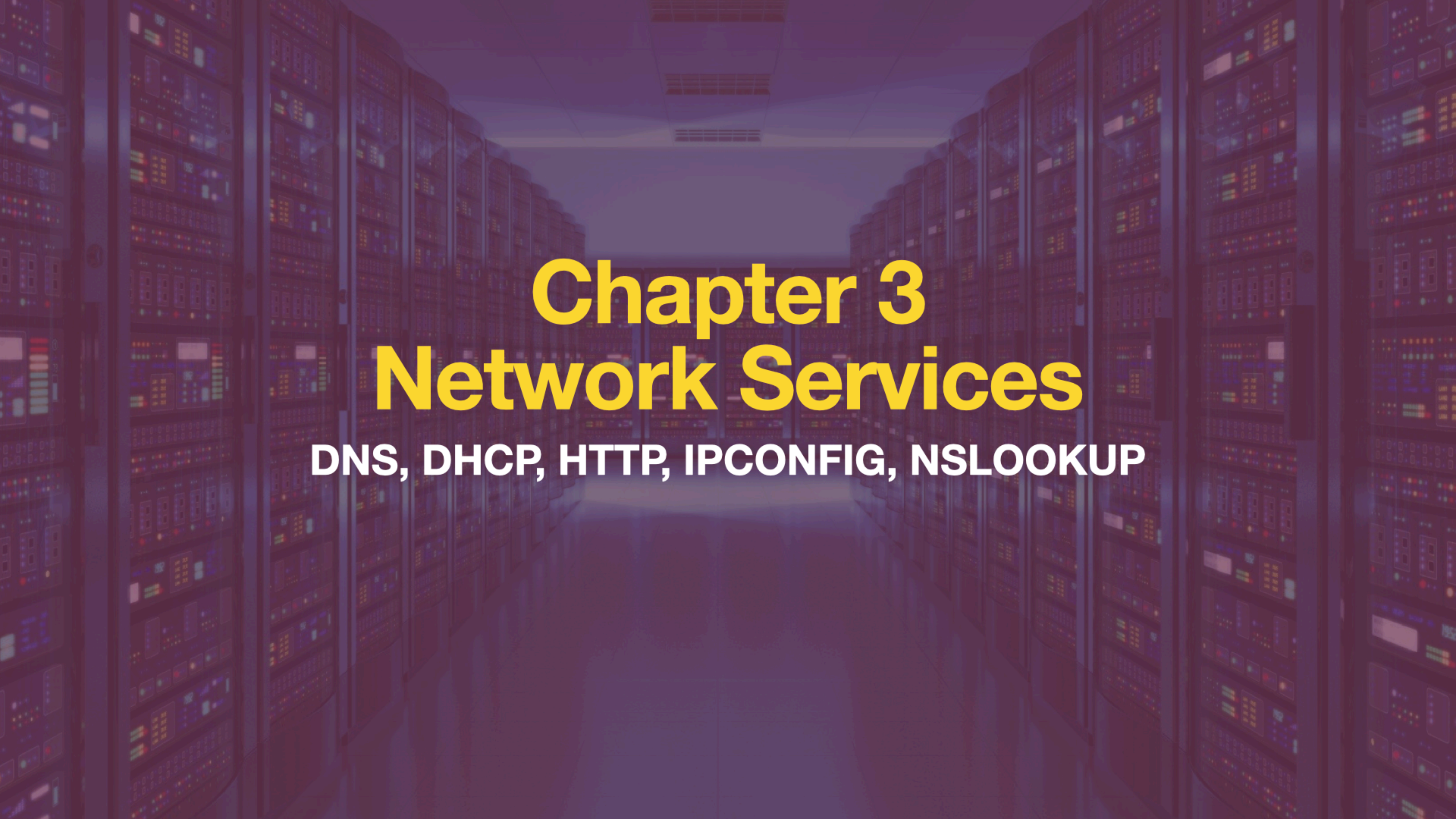
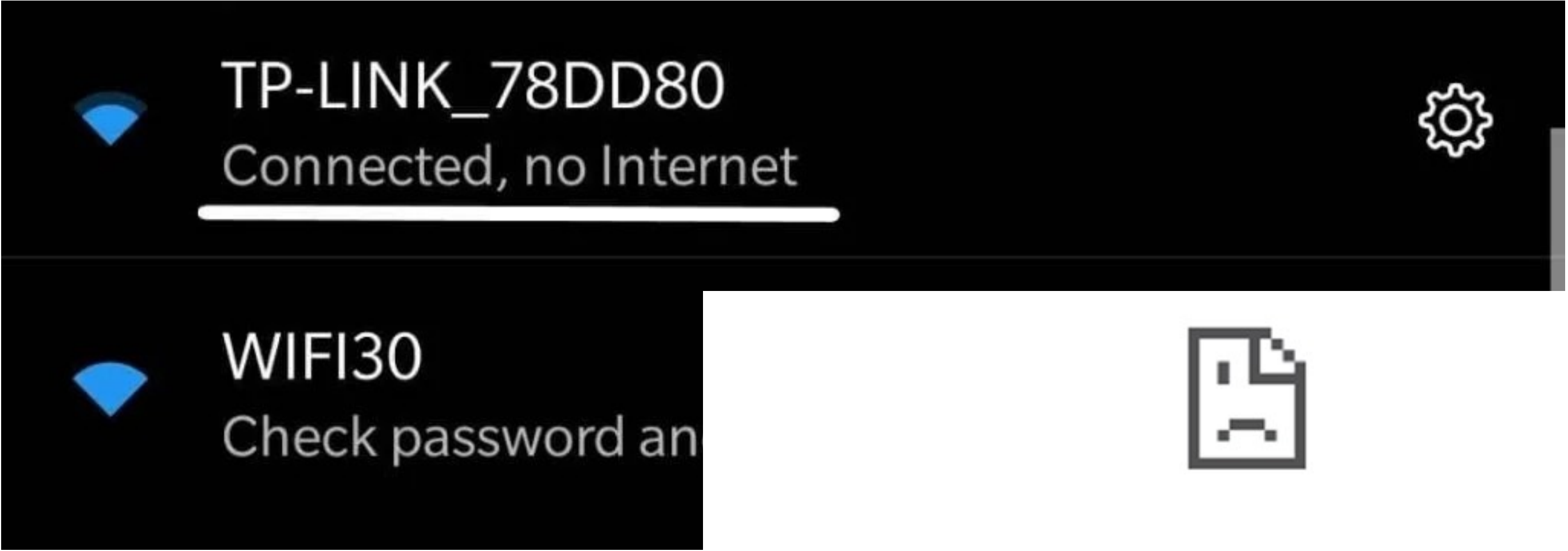




Chapter 3

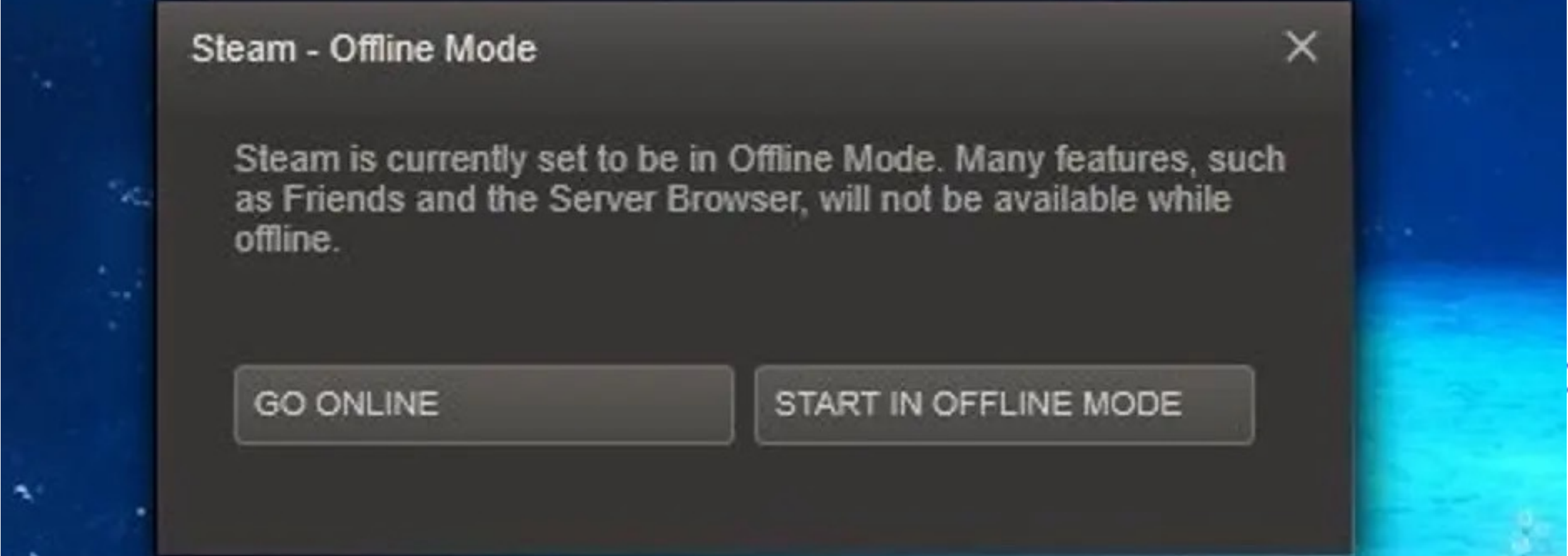
Network Services

DNS, DHCP, HTTP, IPCONFIG, NSLOOKUP

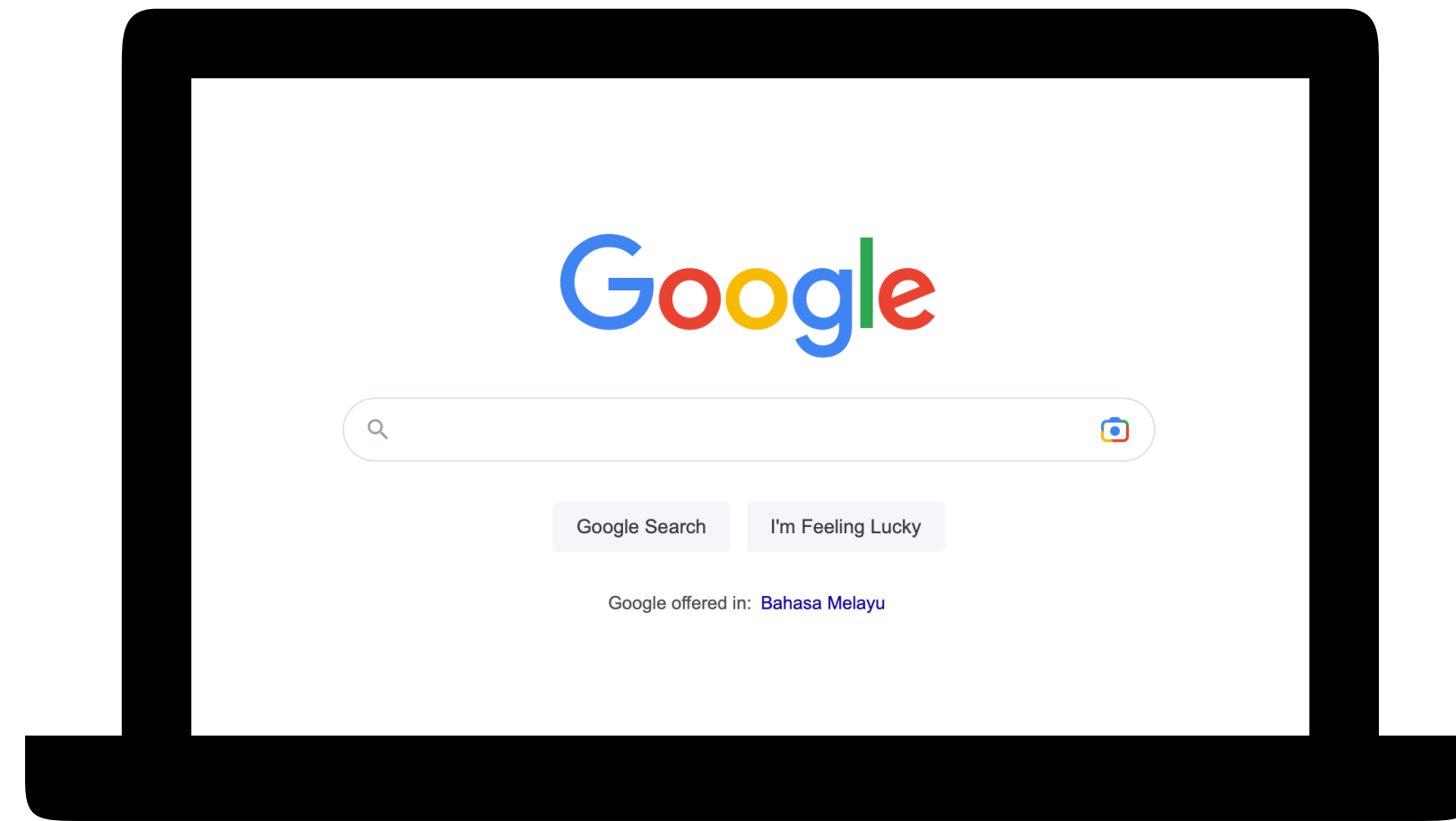


This site can't be reached

www.google.com took too long to respond.



connection
proxy and the firewall
Windows Network Diagnostics



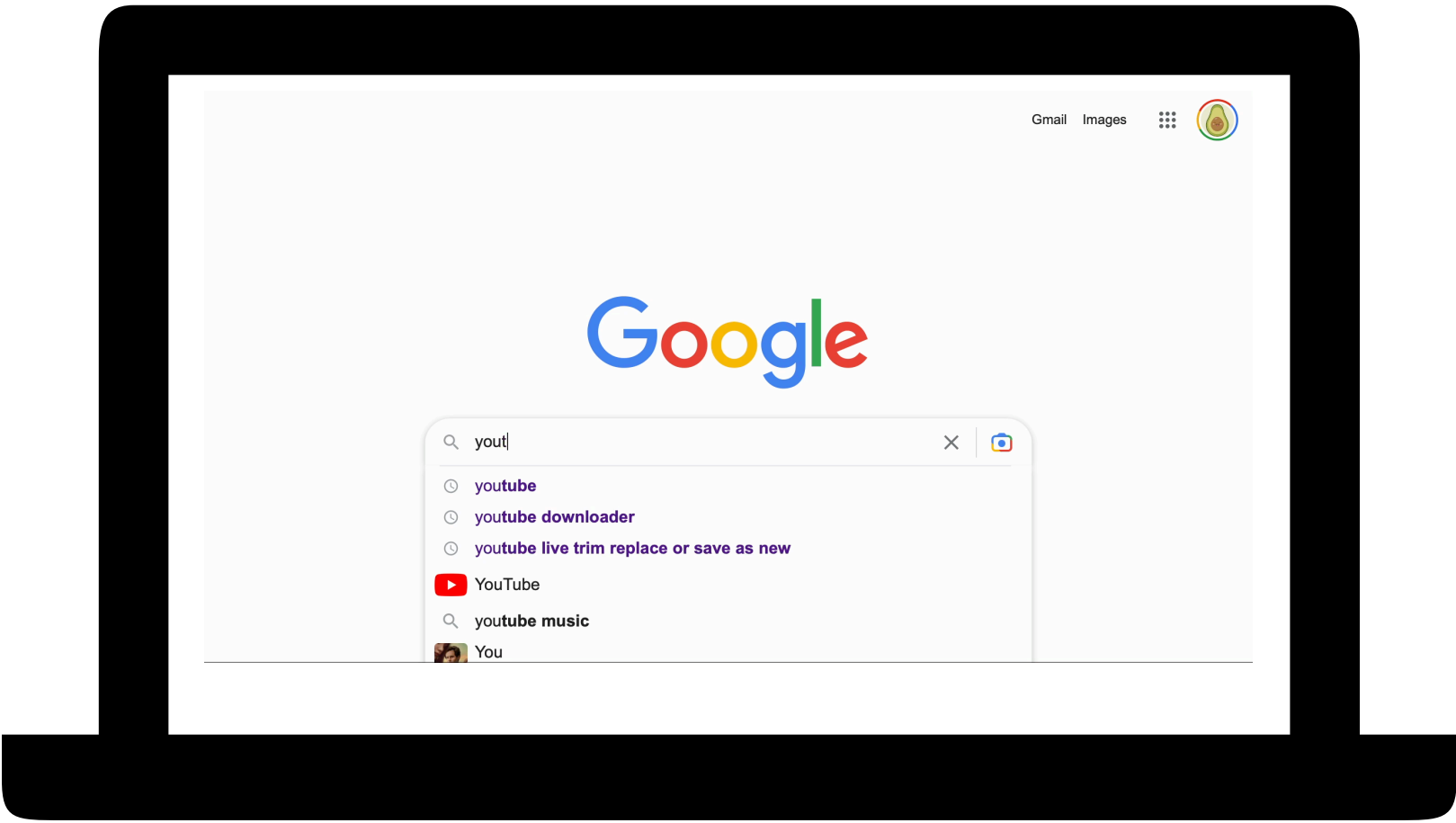
Get IP address
from DHCP Server

Translate
URL to IP
using DNS

Browser send HTTP requests
to Web Server

Browser renders
HTML files





192.168.1.1



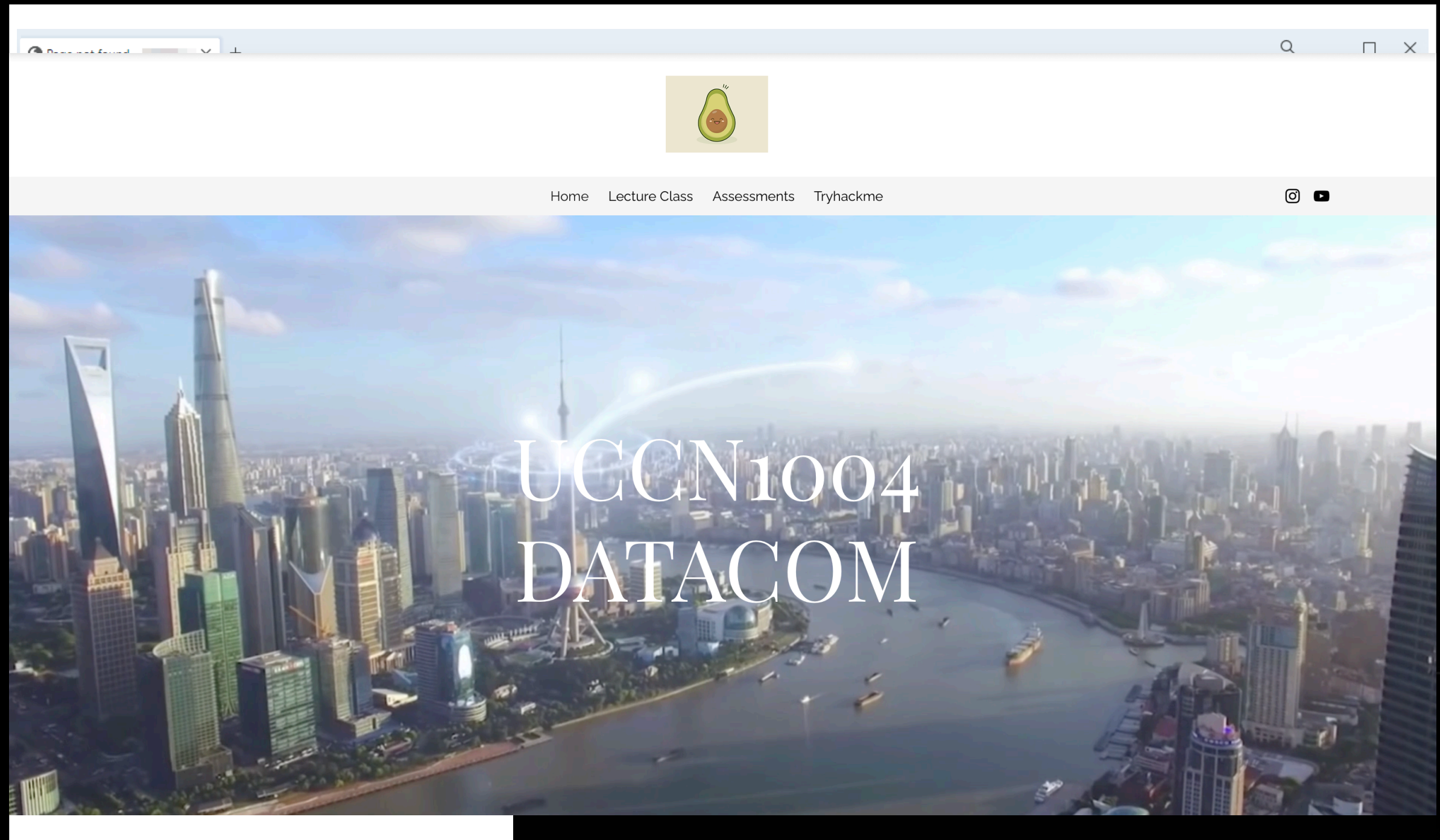
Get IP address
from DHCP Server

Translate
URL to IP
using DNS

Browser send HTTP requests
to Web Server

Browser renders
HTML files





DHCP

DNS

HTTP

Dynamic Host Configuration Protocol (DHCP)



DHCP SERVER

DHCP



POKÉMON

DHCP

Provides dynamic IP address to hosts in a LAN

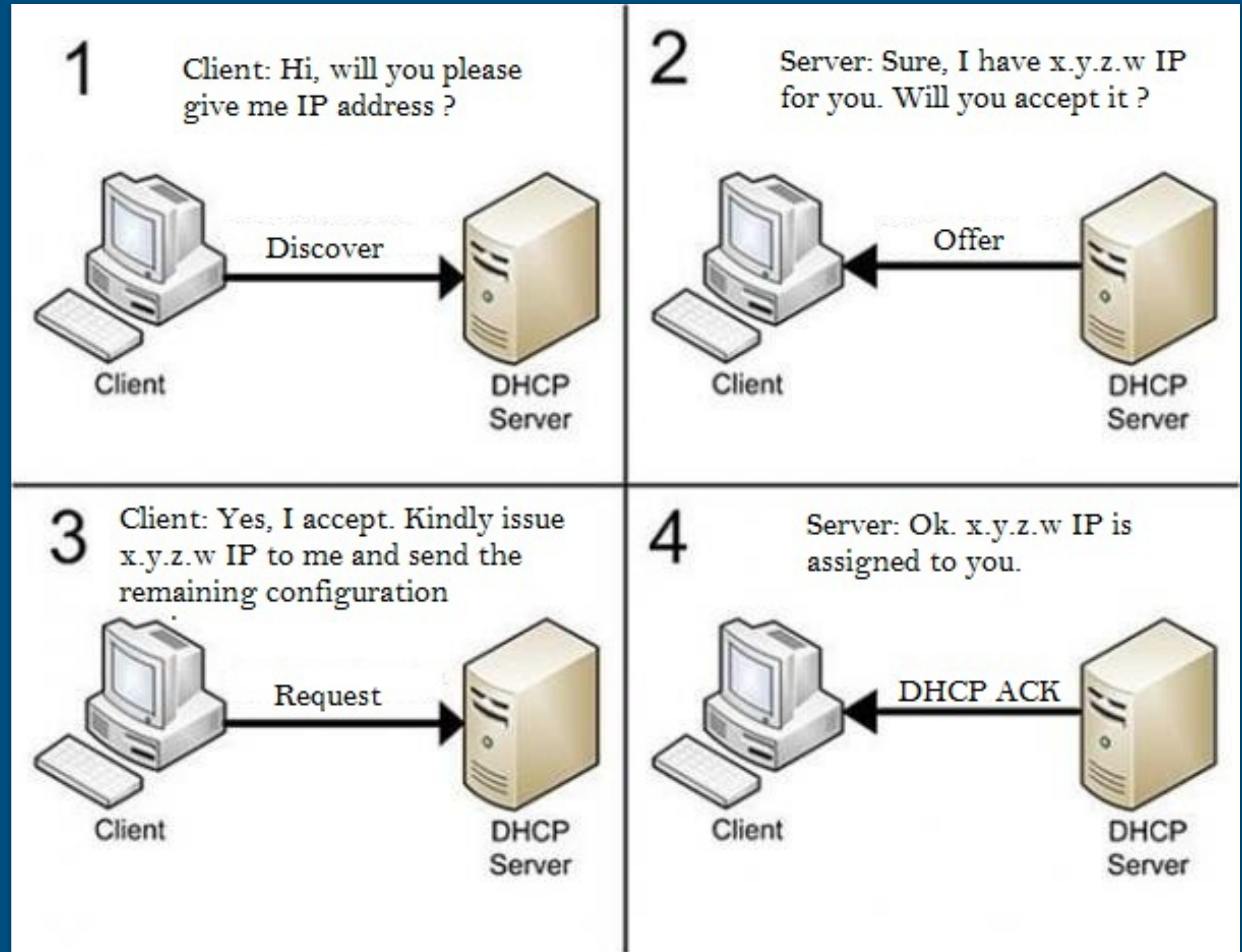
- ANY hosts in a LAN need an IP address to communicate. IP can be static or dynamic.
- A DHCP server automatically assigns a dynamic IP address to the newly connected hosts.



D O R A
I S C O V E R
F F E R
E Q U E S T
C K N O W L E D G E

DHCP DORA

- Using the DORA commands
 - Discover
 - Offer
 - Request
 - Acknowledge



(1) **Discover:**

When a device connects to a network, it sends a broadcast message called a DHCP discover packet to request its configuration parameters. This packet is sent to the broadcast address of the local subnet and is received by all devices on the subnet, including the DHCP server.

(2) **Offer:**

If the DHCP server receives the discover packet, it responds with a DHCP offer packet that includes an available IP address and other configuration parameters. The offer packet is unicast (sent to a specific device) to the requesting device.

(3) **Request:**

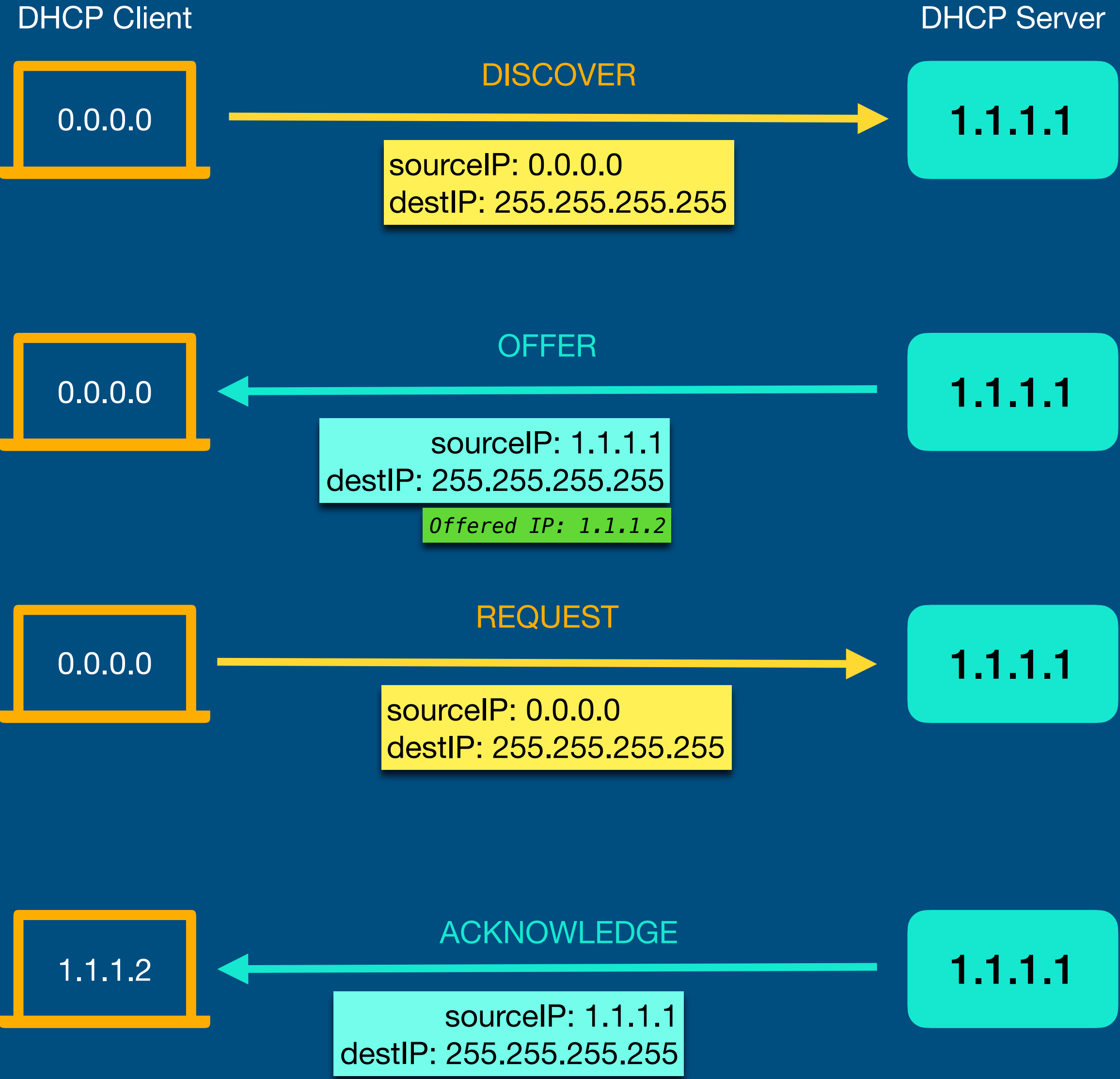
The requesting device responds with a DHCP request packet, acknowledging the offer and requesting that the IP address and other parameters be assigned to it.

(4) **Acknowledge:**

If the DHCP server receives the request, it sends a DHCP acknowledge packet to confirm the assignment of the IP address and other parameters to the requesting device. The device is now configured and ready to communicate on the network.

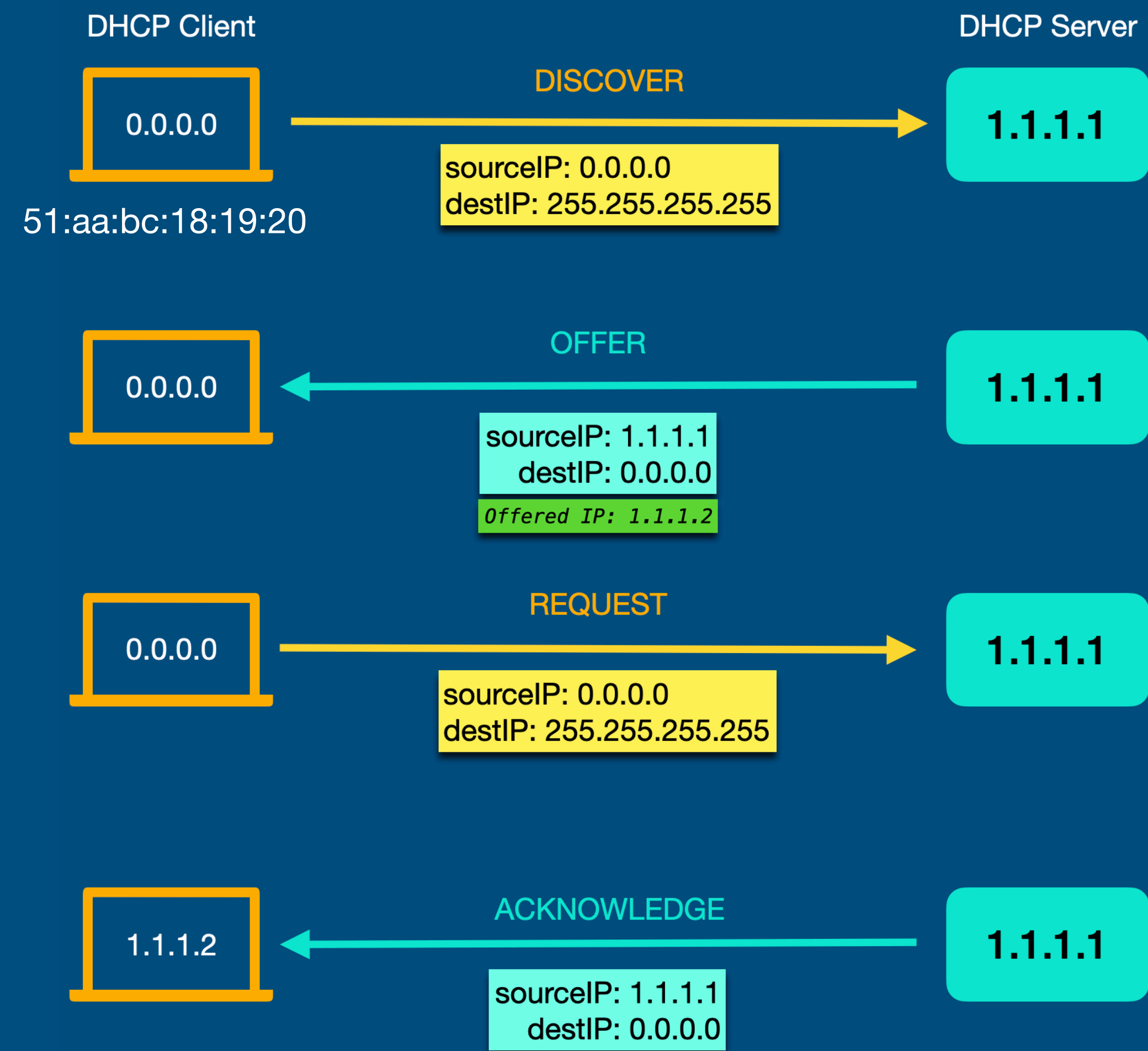
DHCP DORA

Steps by steps



IP-MAC DHCP Binding

How DHCP server always assign the same IP address to the same PC?



DCHP Server (1.1.1.1)

IP Address	MAC address
1.1.1.2	51:aa:bc:18:19:20
1.1.1.3	11:22:ff:33:23:89:19
1.1.1.4	na

IP-MAC Address Binding (Cache)

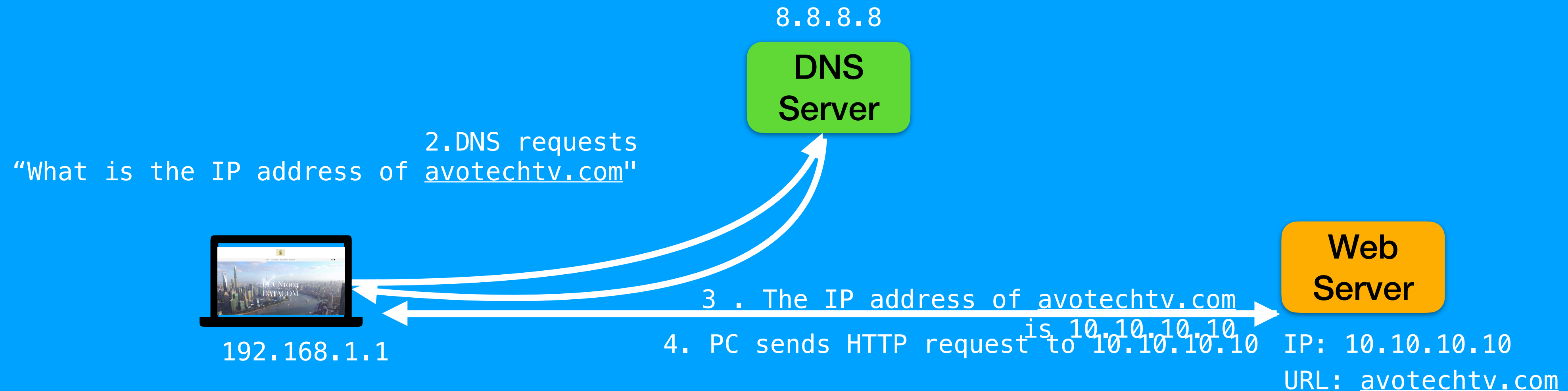
DHCP DEMO

(Packet Tracer)

Domain Name Service (DNS)

Domain Name Service

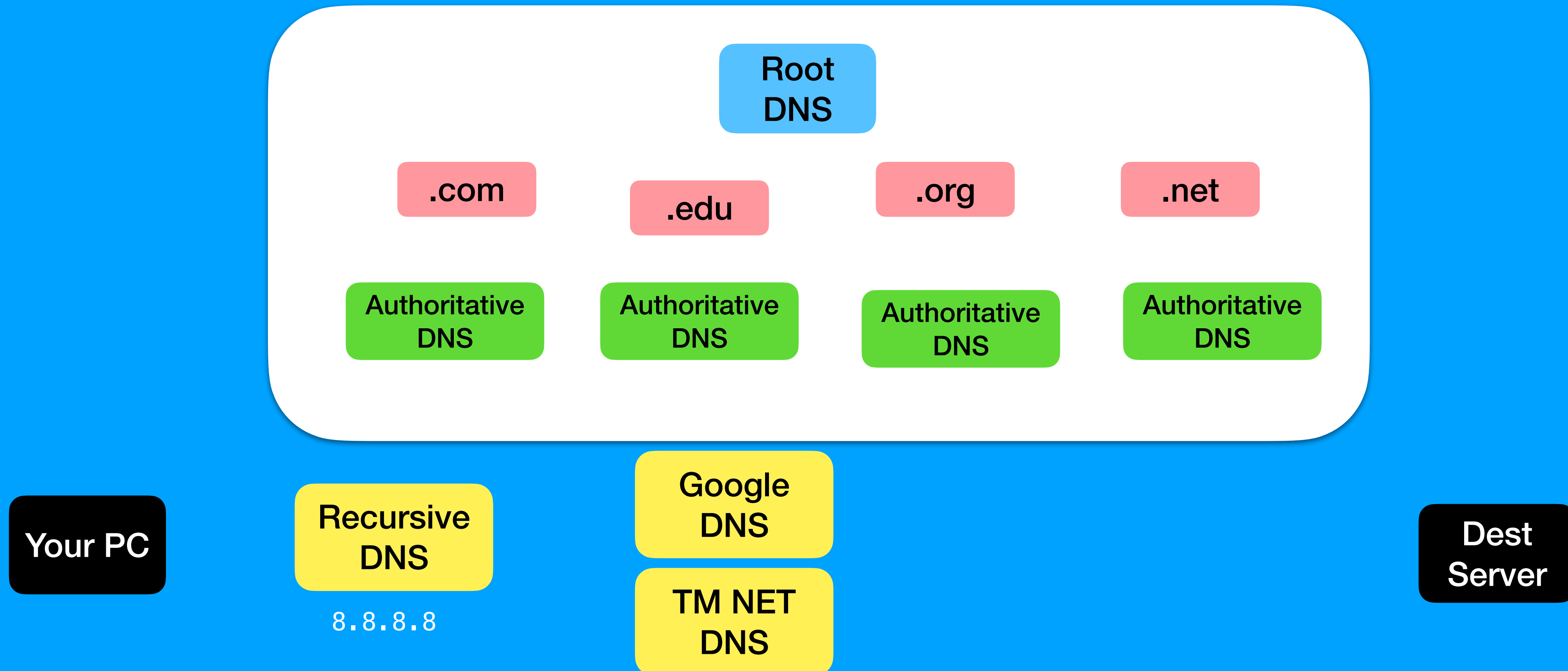
- To translate Web URL (domain names) into IP address



1. User browse to avotechtv.com

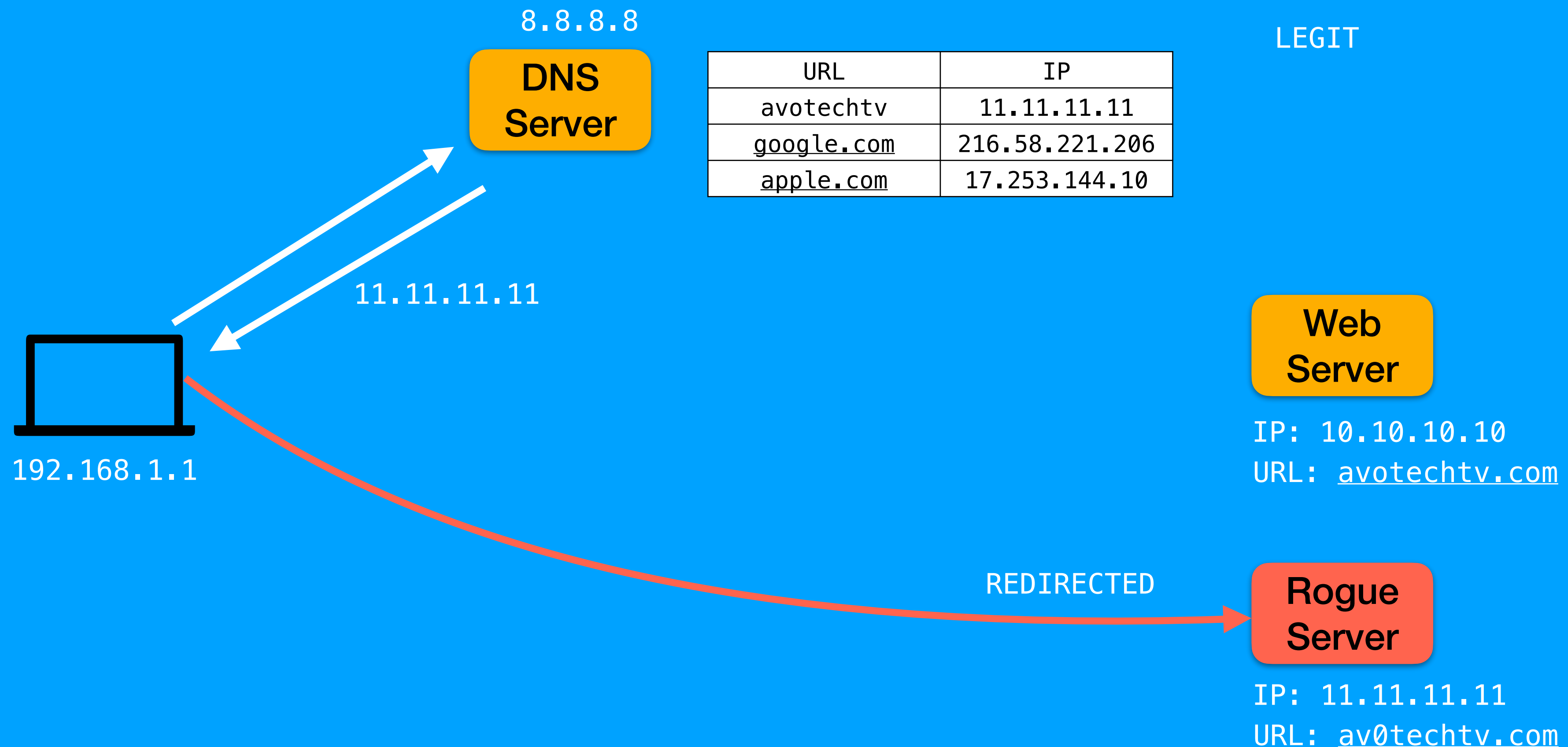
Types of DNS Servers

- Recursive DNS - accept DNS queries, reply with cached records
 - **Root DNS - knows all the 'location' of TLD Server**
 - Top Level Domain (TLD) - contains the 'location' of authoritative (AA) DNS
 - **Authoritative DNS - contains the IP address of the URL being looked up**
- **Order: Recursive -> Root -> TLD -> Authoritative -> Recursive -> Client PC**



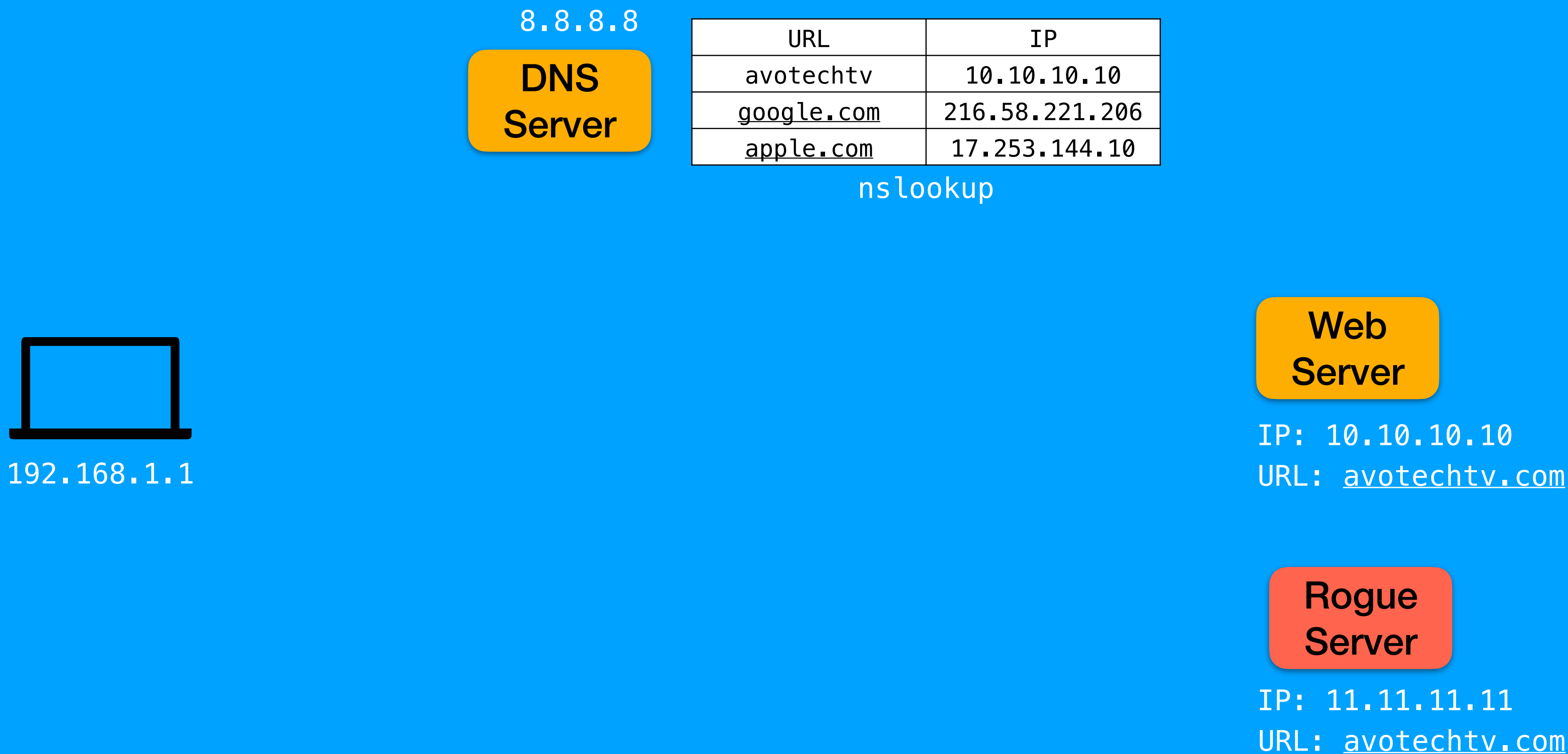
DNS Poisoning

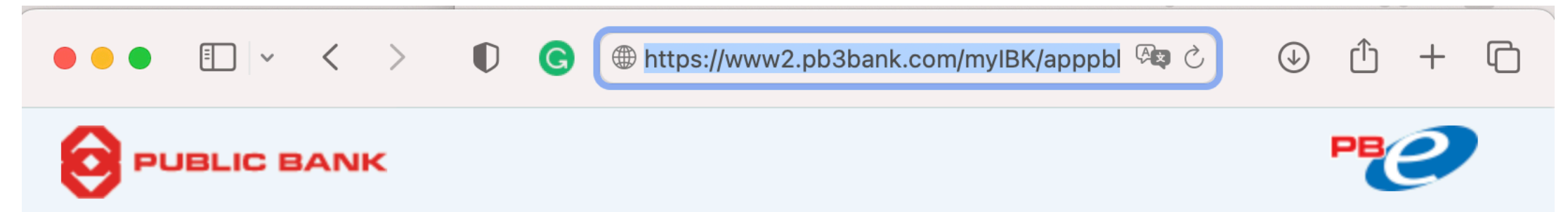
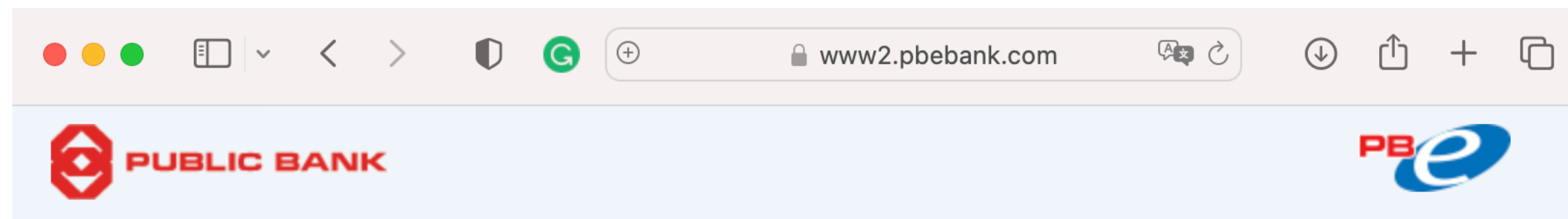
- DNS request/response in cleartext
- DNS records can be modified by attackers during transit
- Attacker can also hack into the DNS servers to change the IP:URL mapping



DNS Poisoning

- DNS request/response in cleartext
- DNS records can be modified by attackers during transit
- Attacker can also hack into the DNS servers to change the IP:URL mapping





Login to PBe

Sunday, 12 February 2023 14:47:18

User ID

Deactivate your online banking access [here](#)

Next

New to PBe?

- ▶ [Register](#)
- ▶ [PBe Online Security](#)
- ▶ [FAQs](#)
- ▶ [PBe Tutorials](#)

Need Help?

- ▶ [Reset Password / Reactivate User ID](#)
- ▶ [Deactivate User ID](#)

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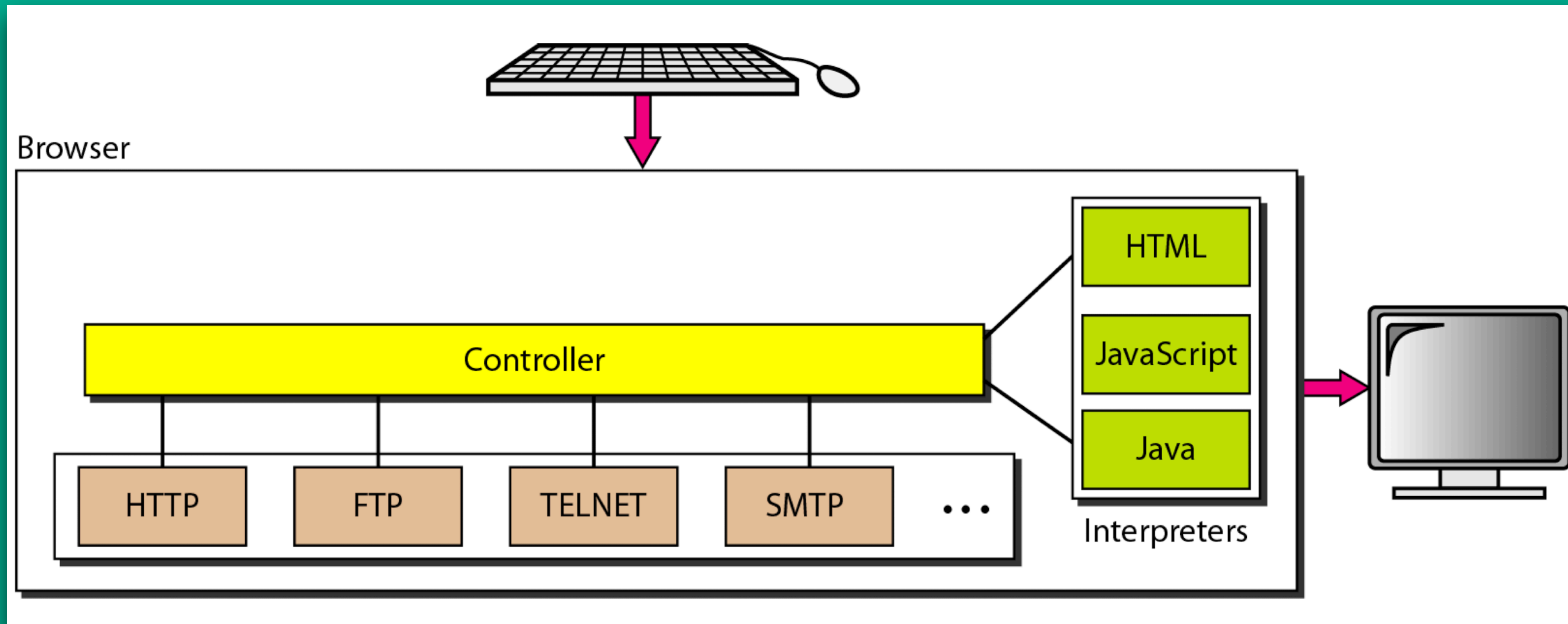
- ▶ [Reset Password / Reactivate User ID](#)
- ▶ [Deactivate User ID](#)

Hypertext Transfer Protocol (HTTP)

Browser

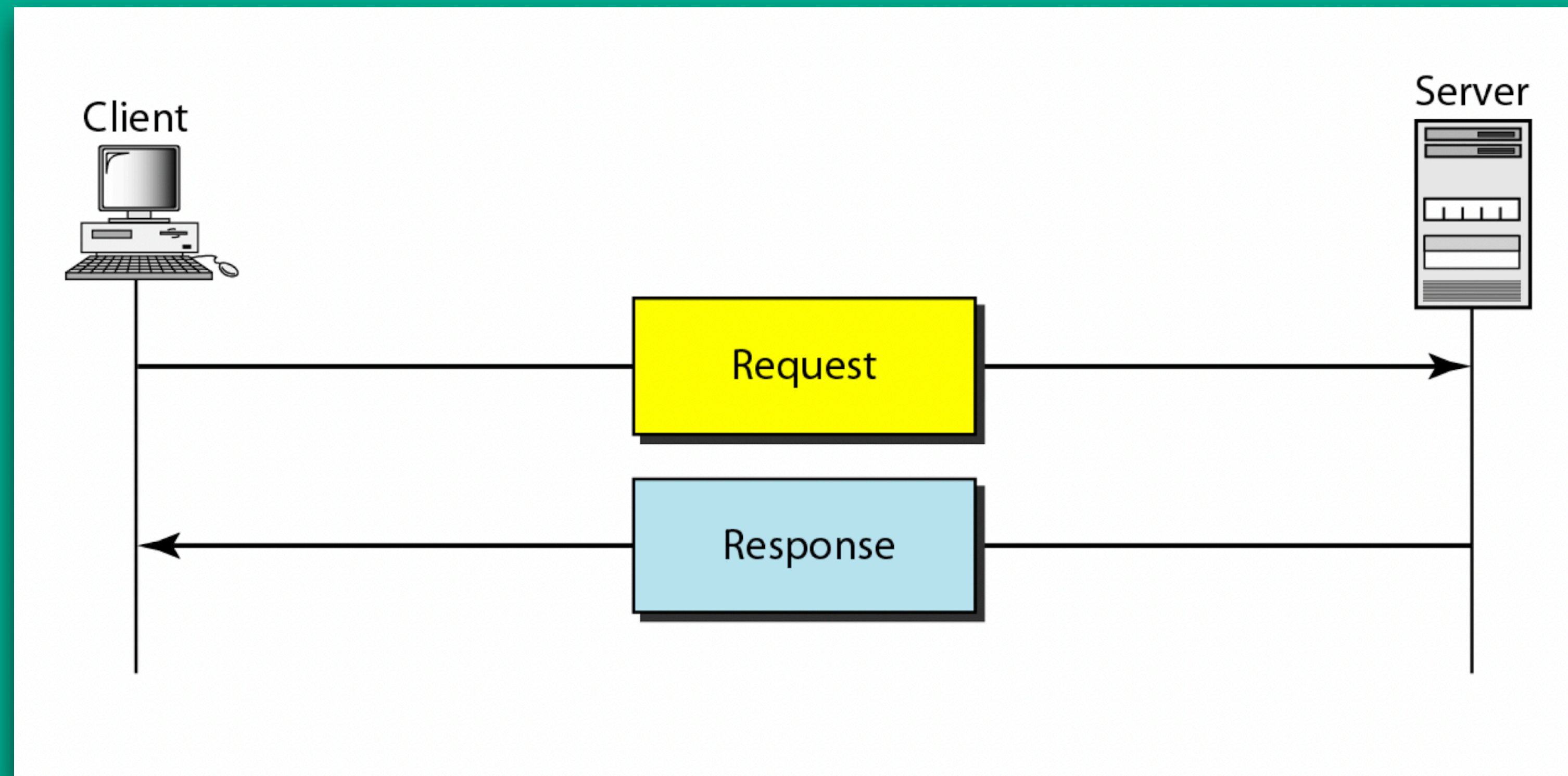


Browser



HTTP

- For sending/receiving hypermedia documents (like HTML)
- **Client** send *http_request*, **Server** reply with *http_response*



HTTP

Client send *http_request*, **Server** reply with *http_response*



HTTP Path



<http://avotechtv.com>

<http://avotechtv.com:8080>

<http://avotechtv.com:8080/main.html>

HTTP Versions

HTTP

- Connections are not secure
- Port 80 or Port 8080
- Vulnerable to Man-in-the-Middle Attack
- Passwords are sent in plaintext

HTTPS

- Connections are secured with SSL/TLS
- HTTP(s), Port 443
- Data sent are encrypted.
- Only the authorised users can see the content.

WEB 1.0, 2.0, 3.0

- Web 1.0 – static webpages, to display information
- Web 2.0 – interactive webpages, content creation, video streaming, browser games
- Web 3.0 – like web2.0, but decentralised

HTML

- HyperText Markup Language (HTML) is the language websites are written in.
- Web browser like Chrome reads HTML document and display the website based on the 'code' in the HTML file.
- Websites are primarily created using:
 - HTML, to build websites and define their structure
 - CSS, to make websites look pretty by adding styling option
 - JavaScript, implement complex features on pages using interactivity

File Transfer Protocol (FTP)



**RM8.90
for 100GB**



**RM3.90 for
50GB**

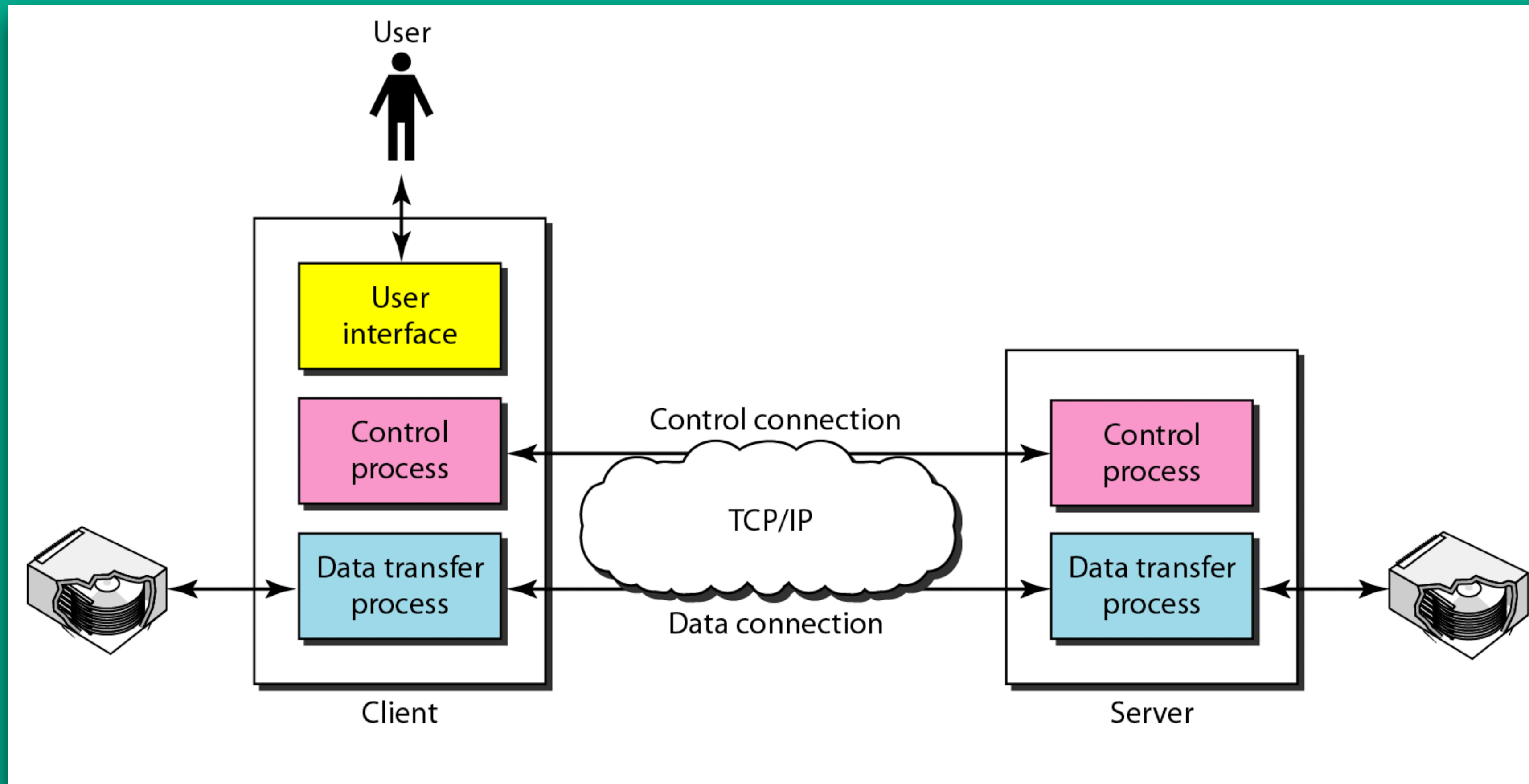
FTP

free*
*NAS
Electricity
Skill*

File Transfer Protocol

Send/receive large data over the Internet

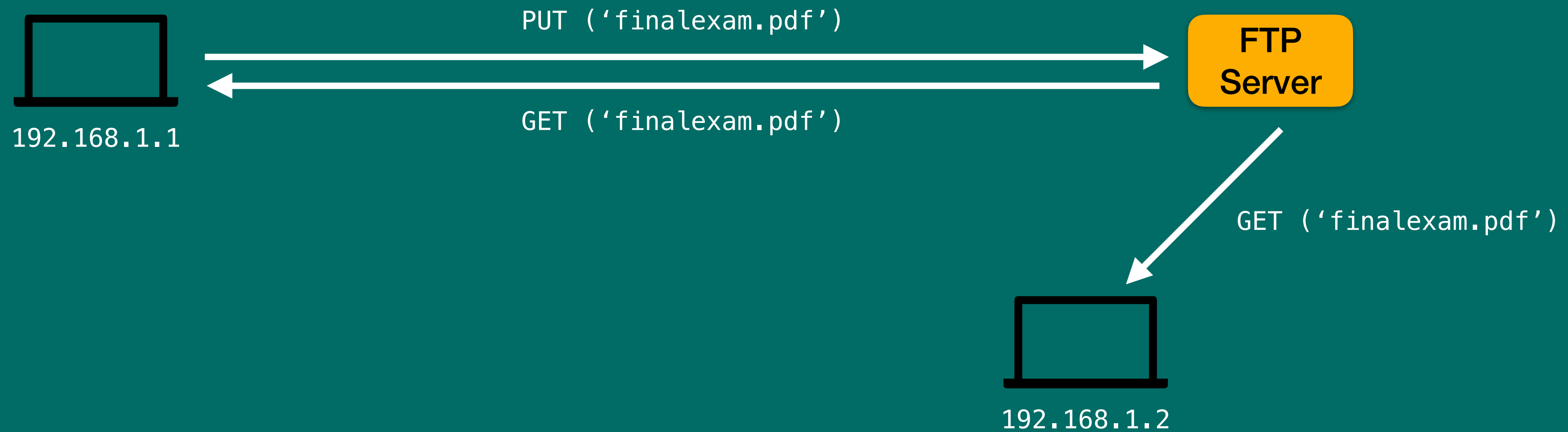
- An FTP server handle data transfer between computers.
- The FTP server waits for clients to connect to it and issue commands that tell the server to upload, download, or list directories.
- FTP runs on TCP/21 (command) and TCP/20 (data)



File Transfer Protocol

Send/receive large data over the Internet

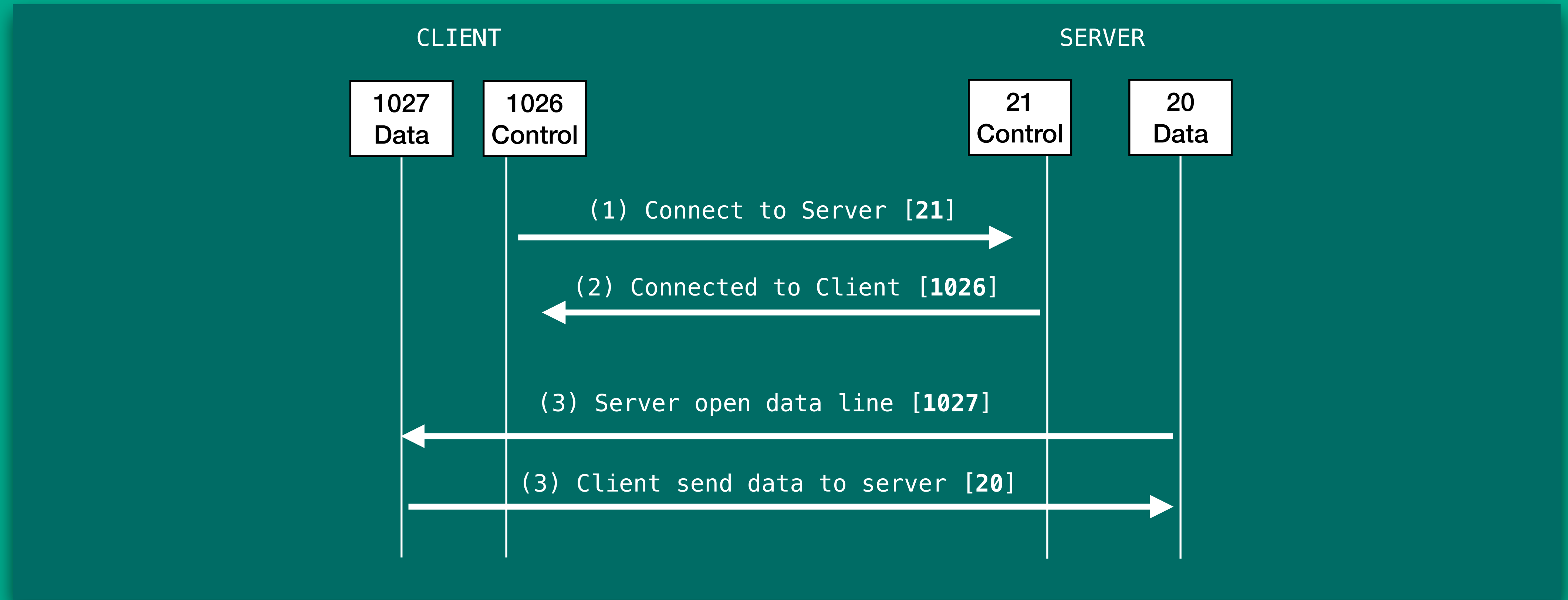
- FTP use commands to upload and download files
- PUT: upload file to server
 - `put finalexam.pdf`
- GET: download file from server
 - `get finalexam.pdf`



Active FTP vs Passive FTP

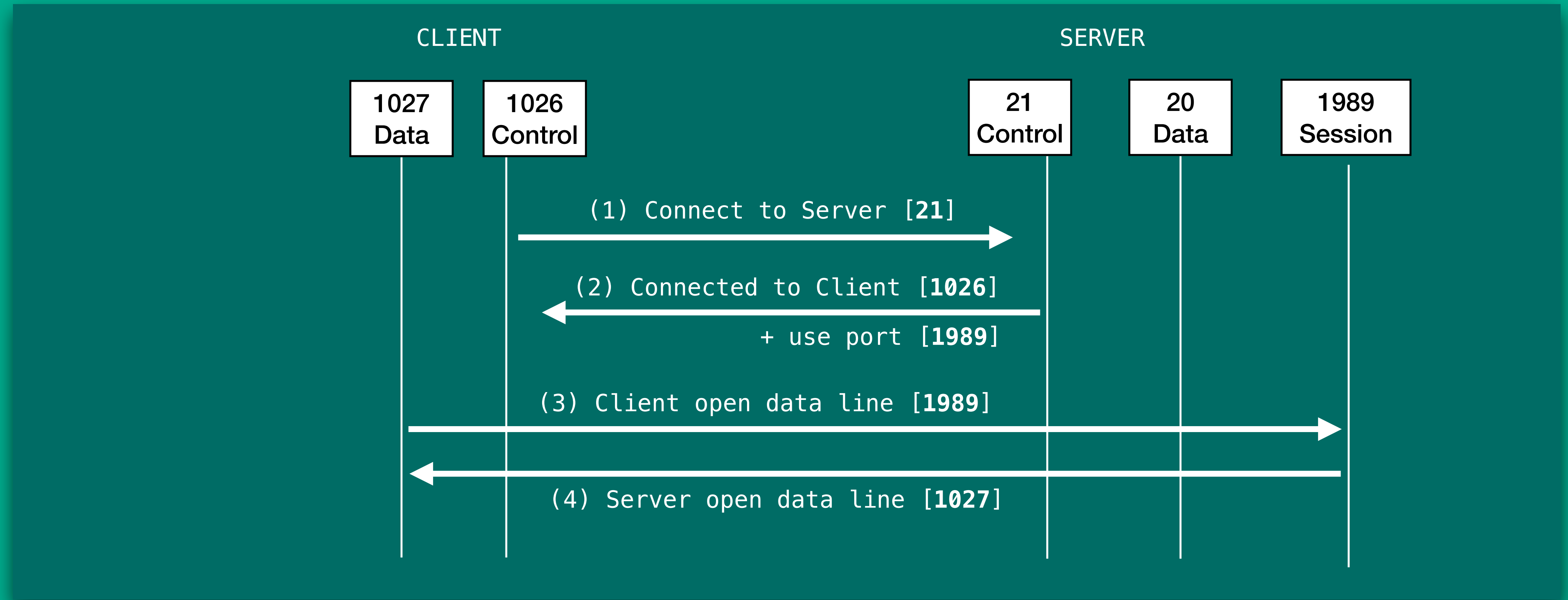
Active FTP	Passive FTP
In active FTP, client establishes the command channel and the server establishes the data channel.	In passive FTP, both the command channel and the data channel are established by the client.
Active FTP provides security to the FTP server.	Passive FTP does not provide security to the FTP server.
Active FTP may cause problems because of firewalls.	Passive FTP does not have connection issues from firewalls.
Active mode is not used as a default mode of a browser.	Passive mode is used as a default mode of a browser.

Active FTP vs Passive FTP



Active FTP

Active FTP vs Passive FTP



PASSIVE FTP

FTP DEMO

(Packet Tracer)